Exploring the Impact of Network Depth on Neural Network Performance - A comparative study between shallow architectures and deep architectures on the problem of image classification

INTRODUCTION

Machine learning models have been widely applied for recognising patterns in data and making predictions without any explicit programming or instructions. Among these models, neural networks have become especially important because of the capability to learn complicated relationships.

One of the major decisions in neural network design is the depth of a neural network, which means the number of layers that are hidden. Increasing the depth of a network can help it learn more complex features, but it can also or instead it can cause issues such as overfitting or the need for more compute time.

This is a tutorial that tries to explore the effects of depth of the neural network, by presenting experimental comparisons between shallow and deep neural network.

CORE CONCEPT

A neural network is a set of several layers connected to each other. These comprise input layer, one or more hidden layers, and output layer. Each layer represents the input data into more and more abstract representations, so that the model can learn the patterns.

The number of hidden layers is what is defined as the depth of a neural network. Shallow networks contain a smaller number of layers whereas deep networks contain multiple layers that can model more complex relations.

Increasing the depth increases the capacity of the model and lets it approximate more complex functions. There are potential issues though, namely:

Underfitting is a term used where a model is too simple to capture the patterns in the data.

Overfitting happens when a model has memorised the data (the training data) and does not generalise (predict in training data).

The instability that occurs in training that can occur in deeper networks when the optimisation is difficult.

Therefore, choosing the right depth is a trade-off between the model complexity and the model generalisation performance.

EXPERIMENTAL SETUP

In order to test the impact of the depth of the network, experiments were performed with the MN model, where images of handwritten digits are found.

Three neural network models were made through TensorFlow:

Model A. Hidden layer (shallow network) 1 hidden layer

Model B: 3 hidden layers

Model C. 5 hidden layers (deep network)

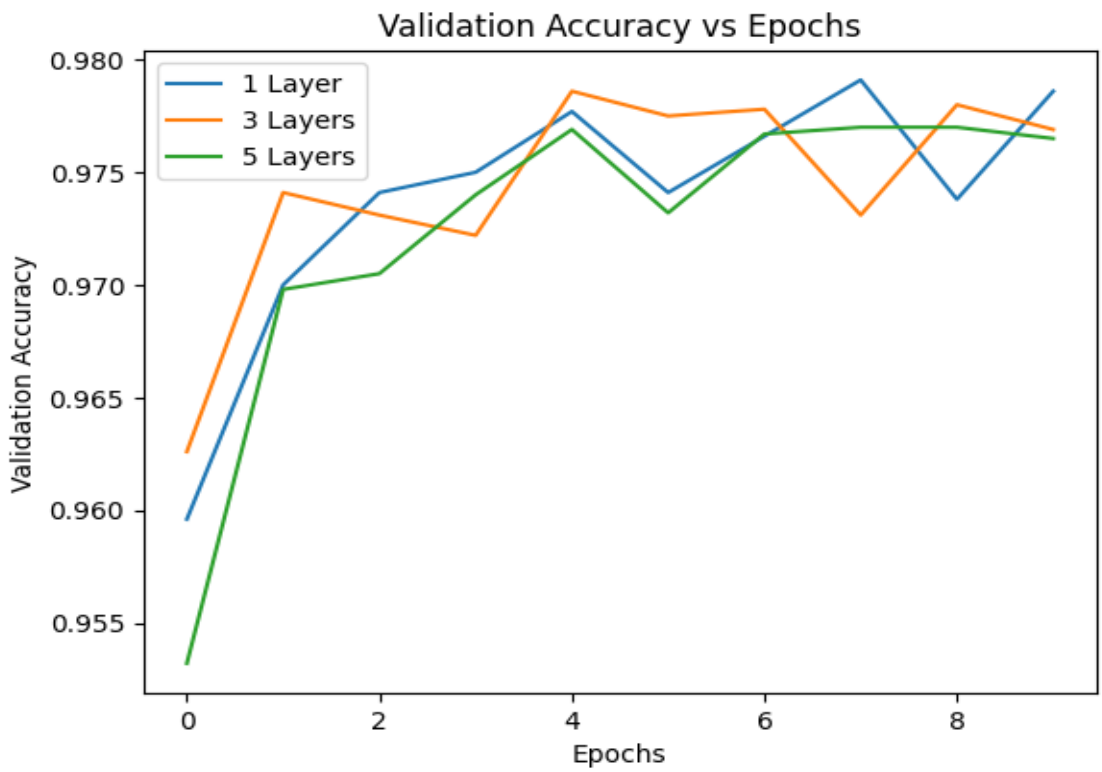
Each of the models had the same number of neurons per layer, and the same activation functions to have a fair comparison between them. The Models were trained for 10 epochs using the Adam optimiser with a categorical cross-entropy loss function.

Performance was measured in terms of validation accuracy/loss.

RESULTS AND VISUALISATION

The analysis of the performance of the models was done using validation accuracy, validation loss, and final result accuracy comparison

Figure 1: Validation Accuracy v/s Epochs for various depths of network

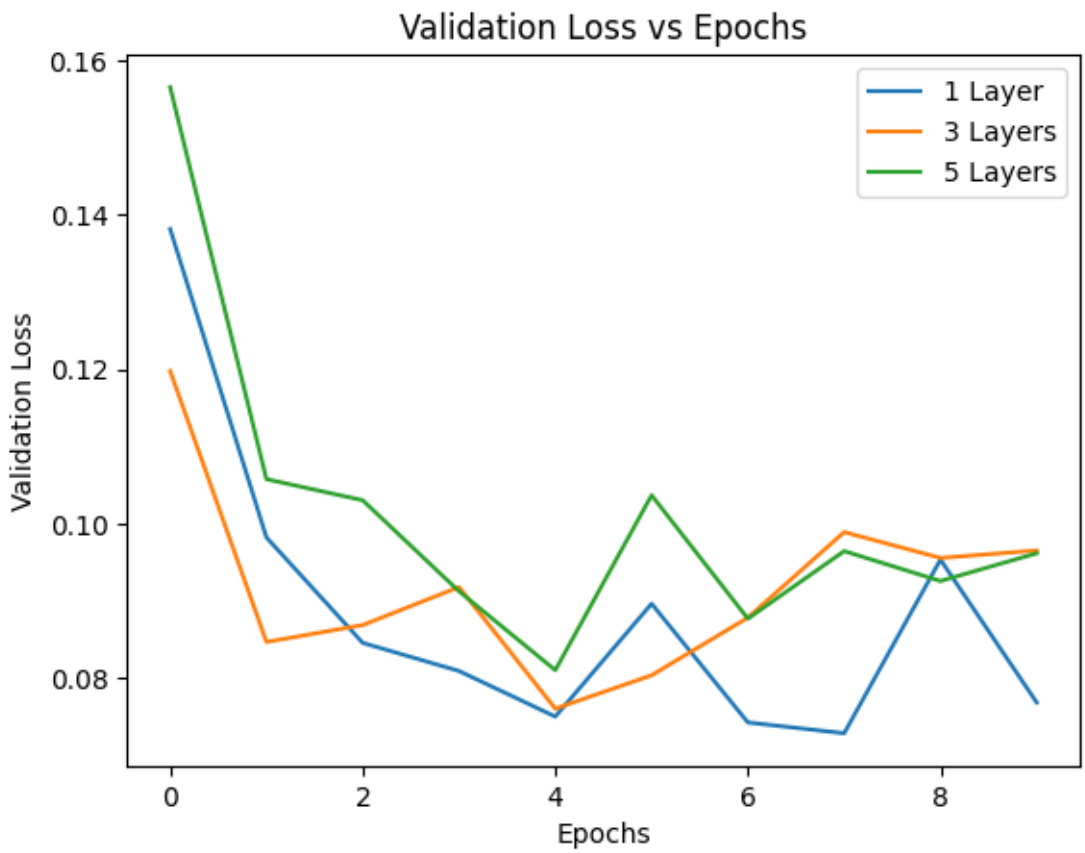


The validation accuracy curves show the improvement of each model over time during the training process. All models show a quick increase in the accuracy throughout the first few epochs, indicating that they are able to quickly grasp features that are useful from the dataset.

The shallow network (1 hidden layer) shows slower improvement, and the accuracy is slightly lower than the deeper ones. In contrast, compared to the three-layer network, the five-layer network and accuracy are achieved much faster, which indicates that the deeper the model, the better the model can discover complex representations.

However, the performance gap between the three-layer and five-layer networks is relatively low, showing that there is little more that can be added in depth to give increase in performance.

Figure 2: Validation Loss vs Epochs for various depths of the network



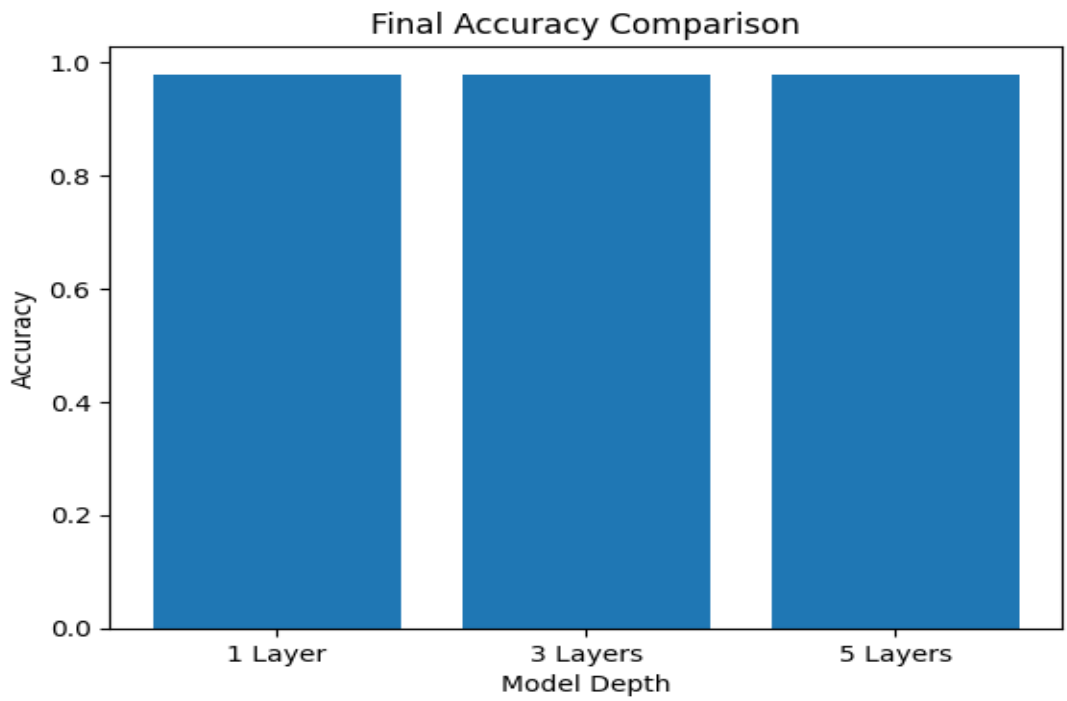
The validation loss curves give us some more insight into the training behaviour of the models. Vibrant models show a drastic drop in loss throughout the first few epochs, indicative of an effective learning.

The shallow network displays relatively stable loss values in later epochs which is a good indicator of generalisation. The three layer model ensures a stable and uniform reduction in loss, indicating balanced learning behaviour.

On the other hand, the validation loss of the five-layer model is more fluctuating, especially at the later epochs. This instability hints at deeper networks being more likely to be prone to overfitting the data or more optimisation, particularly if the dataset is relatively simple.

As we can see in Figure 1 and Figure 2, in deeper models the deeper it is, the more variable the loss becomes which indicates how stable the training is.

Figure 3: Comparison of Form final accuracy among models



The last accuracy comparison makes the overall performance of each model after training. All the three models have high accuracy with values of nearly 97-98%.

Despite the enlarged complexity of deeper networks, the improvement of accuracy is minimal comparing to the shallow model. This means that in the case of the MNIST dataset, which is relatively simple, it makes little sense to increase the depth of networks as performance does not increase significantly.

DISCUSSION

In this experiment, it was found that just above 3 layers didn't increase performance significantly. The results show that deepening the network helps the model to learn complicated patterns, especially in the early stages of training. However, the benefits from increased depth are limited up to a maximum depth.

The three layer network showed the best combination between performance and stability, as it achieved good accuracy while having quite smooth learning behaviour. The five-layer network,

while more complicated, simply did not lead to significant improvements nor did it display indicators of stability.

In this experiment it was observed that the addition of more hidden layers beyond three did not show significant improvements in the accuracy despite increasing the complexity of the model.

This makes it a guess that such deep architectures aren't always required for simpler datasets like MNIST. Instead, balancing the complexity of a particular level is essential to achieving optimal performance.

One local reason for this potentially is that deeper networks will be more challenging to optimise and may experience problems like vanishing gradients, especially with relatively simple datasets. These findings highlight that the design of the model should be based on the complexity of the problem and not on the idea that the more complex model the better.

This behaviour demonstrates the concept of diminishing returns in deep learning, where a greater number of layers will increase the capacity of a model, however this will not necessarily result in any increase in generalisation.

LIMITATIONS AND Theological/Ethical Issues

While deeper neural networks have the ability to improve performance, they can be more computationally intensive and take longer to train. This may restrict their applicability to real-world usage.

Additionally, neural networks are highly dependent on the quality of the training data. Bias in the datasets has the potential to generate unfair or inaccurate predictions, which is therefore a crucial factor to consider when having an ethical approach to machine learning.

CONCLUSION

This study has explored the effect of network depth on network performance. The results showed that as depth increases, so does the ability to learn up to a point, but then there are minimal increases in performance. The results illustrate the importance of a suitable choice of network depth depending on the complexity of the problem. Moderate-depth networks can often offer the best balance between performance and efficiency.

References

- Goodfellow, I., Bengio, Y., & Courville, A. (2016). *Deep Learning*. MIT Press.

- LeCun, Y. et al. (1998). The MNIST Database of Handwritten Digits.
- TensorFlow Documentation**.** **https://www.tensorflow.org**

GitHub:

https://github.com/ekanth3128/Exploring-the-Impact-of-Network-Depth-on-Neural-Network-Performance-/blob/main/Neural%20Depth%20Network.pdf